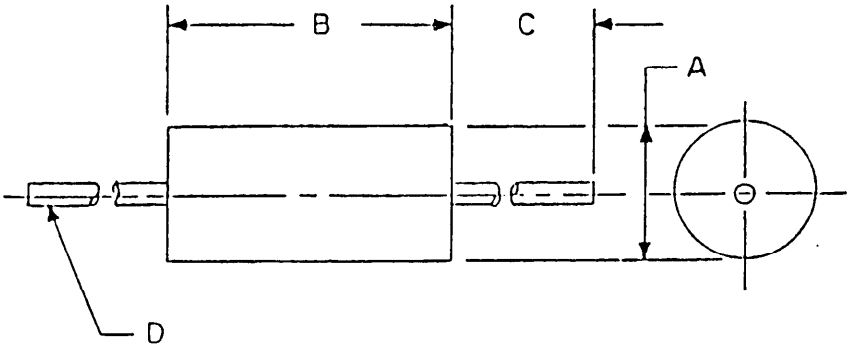
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE, MAGNETICALLY SHIELDED (IRON CORE-IRON SLEEVE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.085 (2.16)	.105 (2.67)
B	.240 (6.10)	.260 (6.60)
C	1.250(31.75)	1.626(41.30)
D	.0185 (.47)	.0215 (.55)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Iron core-iron sleeve.

Weight: 0.25 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 200 volts rms.

Barometric pressure: Method 105 of MIL-STD-202, test condition C, (70,000 feet), test voltage 80 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

Percent coupling: 3 percent maximum.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 3 pounds.

Part number: M39010/13 (dash number from table I).

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number M39010/13 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz) 2/	DC resistance (25°C) max (ohms)	Rated dc current (mA) 3/
AR10**	.10	5, 10	54	25.0	490	.10	670
AR11**	.11	5	52	25.0	430	.11	635
AR12**	.12	5, 10	52	25.0	430	.11	635
AR13**	.13	5	50	25.0	415	.12	610
AR15**	.15	5, 10	50	25.0	415	.12	610
AR16**	.16	5	49	25.0	375	.13	585
AR18**	.18	5, 10	49	25.0	375	.13	585
AR20**	.20	5	47	25.0	330	.15	545
AR22**	.22	5, 10	47	25.0	330	.15	545
AR24**	.24	5	46	25.0	300	.16	530
AR27**	.27	5, 10	46	25.0	300	.16	530
AR30**	.30	5	44	25.0	260	.18	495
AR33**	.33	5, 10	44	25.0	260	.18	495
AR36**	.36	5	42	25.0	230	.19	485
AR39**	.39	5, 10	42	25.0	230	.19	485
AR43**	.43	5	41	25.0	220	.21	460
AR47**	.47	5, 10	41	25.0	220	.21	460
AR51**	.51	5	41	25.0	210	.23	440
AR56**	.56	5, 10	41	25.0	210	.23	440
AR62**	.62	5	39	25.0	180	.24	430
AR68**	.68	5, 10	39	25.0	180	.24	430
AR75**	.75	5	38	25.0	165	.27	405
AR82**	.82	5, 10	38	25.0	165	.27	405
AR91**	.91	5	37	25.0	150	.30	385

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number M39010/13 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz) 2/	DC resistance (25°C) max (ohms)	Rated dc current (mA) 3/
A1R0**	1.0	5, 10	37	25.0	150	.30	385
A1R1**	1.1	5	40	7.9	130	.73	247
A1R2**	1.2	5, 10	40	7.9	130	.73	247
A1R3**	1.3	5	41	7.9	115	.86	228
A1R5**	1.5	5, 10	41	7.9	115	.86	228
A1R6**	1.6	5	43	7.9	105	.95	217
A1R8**	1.8	5, 10	43	7.9	105	.95	217
A2R0**	2.0	5	45	7.9	95	1.10	202
A2R2**	2.2	5, 10	45	7.9	95	1.10	202
A2R4**	2.4	5	48	7.9	90	1.20	193
A2R7**	2.7	5, 10	48	7.9	90	1.20	193
A3R0**	3.0	5	49	7.9	80	1.30	185
A3R3**	3.3	5, 10	49	7.9	80	1.30	185
A3R6**	3.6	5	50	7.9	75	1.50	173
A3R9**	3.9	5, 10	50	7.9	75	1.50	173
A4R3**	4.3	5	53	7.9	70	2.4	136
A4R7**	4.7	5, 10	53	7.9	70	2.4	136
A5R1**	5.1	5	54	7.9	60	2.9	124
A5R6**	5.6	5, 10	54	7.9	60	2.9	124
A6R2**	6.2	5	55	7.9	55	3.2	118
A6R8**	6.8	5, 10	55	7.9	55	3.2	118
A7R5**	7.5	5	55	7.9	53	3.6	111
A8R2**	8.2	5, 10	55	7.9	53	3.6	111
A9R1**	9.1	5	57	7.9	50	4.0	106

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number M39010/13 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz) 2/	DC resistance (25°C) max (ohms)	Rated dc current (mA) 3/
A100**	10.0	5, 10	57	7.9	50	4.0	106
A110**	11.0	5	36	2.5	35	3.0	122
A120**	12.0	5, 10	36	2.5	35	3.0	122
A130**	13.0	5	38	2.5	30	3.4	115
A150**	15.0	5, 10	38	2.5	30	3.4	115
A160**	16.0	5	40	2.5	26	3.8	108
A180**	18.0	5, 10	40	2.5	26	3.8	108
A200**	20.0	5	40	2.5	24	4.9	96
A220**	22.0	5, 10	40	2.5	24	4.9	96
A240**	24.0	5	40	2.5	21	5.8	88
A270**	27.0	5, 10	40	2.5	21	5.8	88
A300**	30.0	5	41	2.5	20	6.5	83
A330**	33.0	5, 10	41	2.5	20	6.5	83
A360**	36.0	5	42	2.5	19	7.9	75
A390**	39.0	5, 10	42	2.5	19	7.9	75
A430**	43.0	5	44	2.5	16	9.3	69
A470**	47.0	5, 10	44	2.5	16	9.3	69
A510**	51.0	5	44	2.5	15	11.0	64
A560**	56.0	5, 10	44	2.5	15	11.0	64
A620**	62.0	5	45	2.5	13	12.0	61
A680**	68.0	5, 10	45	2.5	13	12.0	61
A750**	75.0	5	45	2.5	11	13.0	59
A820**	82.0	5, 10	45	2.5	11	13.0	59
A910**	91.0	5	40	2.5	10.5	16.8	51
A101**	100.0	5, 10	40	2.5	10.5	16.8	51

1/ The complete dash number will include additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.
2/ The SRF minimum values tabulated exceeding 250 MHz are estimates and to be used for reference only.
3/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±5	---	---	-10
Group IV	±10	±(3% +.001 ohm)	1/	-20
Group V	±5	±(3% +.001 ohm)	1/	-10
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(3% +.001 ohm)	1/	-20
Subgroup 3	±5	---	---	-10
Subgroup 4	±10	±(3% +.001 ohm)	1/	-10

1/ The self-resonant frequency shall be not less than the value specified in table I.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 11, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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